LMM-VQA: Advancing Video Quality Assessment With Large Multimodal Models

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***Abstract*—The explosive growth of videos on streaming media platforms has underscored the urgent need for effective video quality assessment (VQA) algorithms to monitor and perceptually optimize the quality of streaming videos. However, VQA remains an extremely challenging task due to the diverse video content and the complex spatial and temporal distortions, thus necessitating more advanced methods to address these issues. Nowadays, large multimodal models (LMMs), such as GPT-4V, have exhibited strong capabilities for various visual understanding tasks, motivating us to leverage the powerful multimodal representation ability of LMMs to solve the VQA task. Therefore, we propose an *L*arge *M*ulti-*M*odal based *V*ideo *Q*uality *A*ssessment (LMM-VQA) model, which introduces a novel spatiotemporal visual modeling strategy for quality-aware feature extraction. Specifically, we reformulate the quality regression problem into a question and answering (Q&A) task and construct Q&A prompts for VQA instruction tuning. Then, we design a spatiotemporal vision encoder to extract spatial and temporal features to represent the quality characteristics of videos, which are subsequently mapped into the language space by the spatiotemporal projector for modality alignment. Finally, the aligned visual tokens and the quality-inquired text tokens are aggregated as inputs for the large language model (LLM) to generate the quality score as well as the quality level. Extensive experiments demonstrate that *LMM-VQA* achieves state-of-the-art performance across five VQA benchmarks, exhibiting an average improvement of 5% in generalization ability over existing methods. Furthermore, due to the advanced design of the spatiotemporal encoder and projector, LMM-VQA also performs exceptionally well on general video understanding tasks, further validating its effectiveness. Our code will be released at https://github.com/Sueqk/LMM-VQA**

***Index Terms*—Video quality assessment (VQA), large multimodal models (LMMs), modality alignment.**

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I. Introduction

As VIDEO consumption continues to surge across a variety of streaming media platforms, including YouTube, TikTok, Facebook, Netflix, etc, accurately evaluating the perceptual quality of videos is crucial for these video-enabled applications and services to monitor video quality throughout the entire video processing procedures and perceptually optimize video procession algorithms (*e.g.* video compression and enhancement), ultimately providing better quality of experience (QoE) for end-users. Towards this goal, several video quality assessment (VQA) methods [42] have been proposed in recent years, which can be categorized into reference-based VQA, including full-reference [2], [57] and reduced-reference VQA [20], [55], and no-reference VQA, also known as blind VQA (BVQA) [26], [27], [28], [50], [56], [58], [59], [62], [63], [65], [67], [69], [75], [76], [85], [86]. In this paper, we focus on BVQA[1] as it does not require any reference video information and has a broader range of application scenarios.

In the literature, BVQA is typically studied from two perspectives: the knowledge-driven approach [26], [43], [62], [63] and the data-driven approach [27], [28], [56], [58], [65], [67], [69], [76], [86]. Knowledge-driven BVQA methods leverage prior knowledge of visual quality perception to extract corresponding handcrafted features for quality regression. For example, natural scene statistics (NSS) [45] have been proven to be sensitive to synthetic distortions and are the most frequently used features in knowledge-driven blind image and video quality assessment studies [43], [44], [49]. Other visual descriptors, such as texture [12], noise [18], contrast [19], blockiness [88], color [52], motion vector [5], optical flow [41], etc., are also considered quality-related features and have been explored in previous studies.

In contrast, data-driven BVQA methods automatically learn the quality-aware feature representation by training a carefully designed deep neural network (DNN) in a learning-based manner. Typically, there are three kinds of architectures for quality-aware feature extraction: 1) utilizing a 2D network to extract frame-wise features and then employing a sequence model (*e.g.* GRU [7]) to fuse them into video-level features [28], [75]; 2) utilizing a 2D network to extract key frame features and a 3D network to extract the motion features from video chunks, which are then fused into video-level features by the concatenating operator or a sequence model [56], [69]; 3) directly utilizing a 3D network to extract video-level features from video chunks [67]. The 2D and 3D back-

[1]In the following paper, we also use VQA to refer to BVQA for simplicity.

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TABLE I

THE PERFORMANCE OF LATEST LMMS ON VARYING VQA DATASETS. WE UTILIZE THE TEXT PROMPTS AND VIDEO AS INPUT AND ASK LMMS TO GENERATE THE SPECIFIC SCORES

Dataset	**$LSVQ_{test}$**		**$LSVQ_{1080p}$**		**KoNViD-1k**		**YouTube-UGC**		**LIVE-VQC**		**LIVE-YT-Gaming**	
Method	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
Video-ChatGPT [40]	0.060	0.055	0.055	0.065	0.010	0.058	0.037	0.054	0.005	0.008	0.051	0.050
LLaMA-VID [31]	0.121	0.133	0.093	0.119	0.145	0.183	0.091	0.118	0.099	0.125	0.023	0.045
VideoLLaMA2 [6]	0.220	0.294	0.129	0.227	0.080	0.142	0.174	0.260	0.090	0.124	0.007	0.058
LLaVA-NeXT-Video [29]	0.288	0.318	0.246	0.274	0.287	0.381	0.287	0.381	0.094	0.128	0.171	0.175
VILA-1.5 [33]	0.470	0.484	0.432	0.508	0.566	0.581	0.412	0.465	0.392	0.461	0.217	0.243

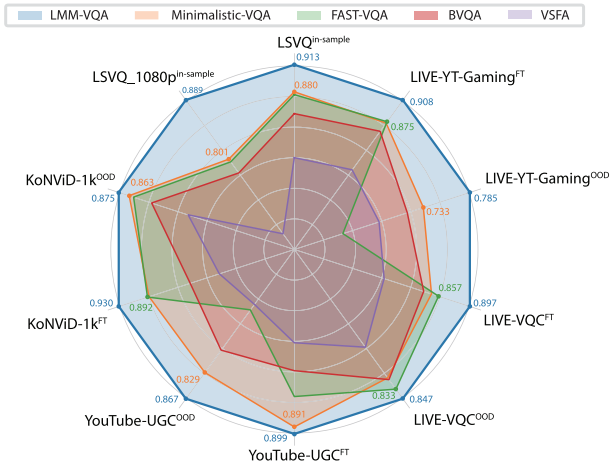

Fig. 1. An overall performance comparison between **LMM-VQA** and existing state-of-the-arts methods. KoNViD-$1k^{OOD}$ indicates that the model is trained on the LSVQ dataset and evaluated on the KoNViD-1k dataset. KoNViD-$1k^{FT}$ refers that the model is pre-trained on the LSVQ dataset, fine-tuned on KoNViD-1k, and evaluated on the KoNViD-1k dataset. $LSVQ^{in\text{-}sample}$ represents that the model is trained on the LSVQ dataset and evaluated on the LSVQ dataset. Metrics are (SRCC+PLCC)/2.

bones can be pre-trained from other computer vision tasks [21], [77] or fine-tuned on VQA datasets via an end-to-end manner.

Despite that significant efforts have been made in the field of BVQA, existing methods still face several challenges. First, knowledge-driven BVQA methods have better explainability but perform poorly on in-the-wild videos, as handcrafted features cannot model the complex in-the-wild distortion diverse video content. Second, although data-driven BVQA methods perform well on specific VQA datasets, they struggle to maintain the same level of performance during out-of-distribution (OOD) evaluations (as shown in Fig. 1), which is more critical for real-world applications. This limitation arises because current VQA datasets contain a relatively small scale of videos, which are insufficient to cover the vast high-dimensional spaces of video content and distortions, thereby restricting the ability of trained BVQA models to generalize to OOD videos.

To overcome this problem, a straightforward approach is to perform subjective experiments to construct a larger-scale VQA dataset that encompasses a wider variety of video samples. However, subjective VQA experiments are highly expensive and time-consuming, making the construction of such a large-scale VQA dataset challenging. Another routine is to leverage as many existing visual understanding models as possible to enhance the feature representation of BVQA models. For example, some BVQA studies [36], [59], [65], [79] combine diverse visual task models and quality assessment models to boost their representation capacity.

TABLE II

THE PROBABILITY OF INVALID ANSWERS ACROSS VARIOUS MODEL STRUCTURES. THE PRE-TRAINED PARAMETERS ARE FROM VIDEO-CHATGPT [40]. W/P AND WO/P RESPECTIVELY INDICATE THAT WE LOAD PRE-TRAINED PARAMETERS AND DO NOT LOAD PRE-TRAINED PARAMETERS FOR TRAINING

Projection layer	**pre-trained weights**	**Invalid answers ↓**
MLP	w/p	59.0%
MLP	wo/p	12.0%

Recently, the emergence of large multimodal models (LMMs) [1], [10], [31], [35], [80] opens the door for vision foundation models to handle general-purpose visual tasks. Some work also reveals that LMMs have strong abilities to understand low-level visual characteristics [34], [70], [71], [81]. These encouraging advances inspire us to leverage LMMs to address the BVQA task, thus benefiting from their strong general-purpose visual representation capabilities. However, our empirical studies reveal that directly utilizing video LMMs as the quality evaluator leads to abysmal performance. As shown in Table I, the SRCC and PLCC values of the best-performing model, VILA-1.5, do not exceed 0.6 on five VQA benchmarks. This limitation arises because these multimodal models are pre-trained on video understanding tasks, which focus on describing video content rather than evaluating its quality. This semantic gap poses challenges in extracting tokens relevant to video quality scores. An additional piece of evidence is the instability of current LMMs in generating consistent video quality scores. As shown in Table II, when using Video-ChatGPT [40] as a baseline model trained on the VQA dataset, incorporating pre-trained weights in the projection layer increases the probability of generating invalid responses from 12% to 59%. This demonstrates that weights pre-trained for video understanding tasks can negatively impact video quality assessment. Therefore, *how to adapt LMMs to the VQA task and leverage their strong visual representation capabilities to develop effective and robust VQA model remains an open challenge*.

Toward this goal, we propose **a large multimodal-based video quality assessment (LMM-VQA) model** by instruction tuning a spatiotemporal enhanced LMM model on the VQA datasets. Given that the functionality of LMMs is defined by instruction prompts, we design a rule-based approach to

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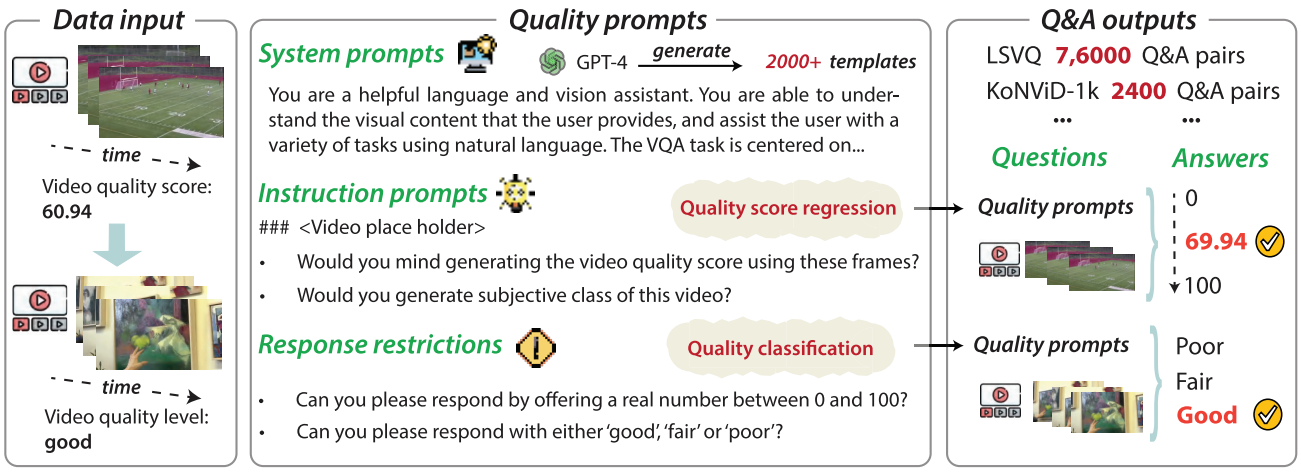

Fig. 2. **The construction of pair-wised Q&A instruction prompts for the training of LMM-VQA**. Given the video input and its quality score, we construct pair-wised Q&A instruction prompts for two tasks: quality score regression and quality classification. We leverage the GPT-4 annotator to generate 2000 templates of quality prompts, which include system prompts, instruction prompts, and response restrictions.

automatically generate a set of question-and-answer (Q&A) pairs for the VQA datasets. As shown in Fig. 2, the question template, termed quality prompts, consists of one system prompt, two instruction prompts, and two response restrictions to ensure the LMM comprehends the VQA task and produces the correct output format. The answering template contains the video quality score and the quality level, serving as the ground truth outputs for fine-tuning the LMM-VQA model.

Our LMM-VQA model consists of three modules: a spatiotemporal enhanced visual encoder for video feature extraction, a spatiotemporal visual project for vision-language alignment, and a large language model (LLM) for understanding video quality characteristics. For the spatiotemporal enhanced visual encoder, we avoid the sparse sampling frame strategy commonly used in most video LMMs [30], [32], [74] to encode video features, as temporal distortions in videos are typically reflected in continuous frames. Instead, we treat the video features as two distinct parts: spatial features and temporal features, where the spatial features are extracted by a 2D visual encoder (*e.g.,* CLIP ViT-L/14 [48]) from sparse key frames, while the temporal features are extracted by a 3D visual encoder (*e.g.,* SlowFast [15]) from continuous frames. This approach empowers LMMs to capture enough spatial and temporal distortion cues in videos for quality analysis, thereby enhancing prediction performance for the VQA task. Subsequently, the spatial and temporal features are inputted to the spatiotemporal projector to map visual tokens into the language space for modality alignment. Finally, the aligned visual tokens and quality prompt tokens are aggregated as input for the LLM to generate the final answers, including the video quality scores and levels. Experimental results reveal that LMM-VQA achieves state-of-the-art (SOTA) performance across different datasets. Specifically, we apply in-sample, OOD, and fine-tuning evaluations to demonstrate its effectiveness in VQA tasks.

Our contributions are summarized as follows:

- We develop an instruction tuning pipeline to adapt LMMs for the VQA task, effectively leveraging their powerful visual representation capabilities for high-performance and robust video quality evaluation.
- To address tricky temporal distortion issues in videos, we propose a spatiotemporal enhanced visual encoder that extracts the spatial and temporal features separately. The temporal features enhance the perception of temporal distortions and significantly improve the performance of the proposed LMM-VQA.
- To map the quality-related video representations into text modality, we design novel spatial and temporal projectors and construct Q&A Instruction prompts for instruction tuning.

II. Related Work

A. Blind Visual Quality Assessment

Knowledge-driven models primarily utilize handcrafted features derived through prior knowledge of visual quality perception to evaluate video quality [26], [43], [62], [63]. For instance, NIQE [45] and BRISQUE [43] first introduce natural scene statistics (NSS) for general quality predictions. Subsequently, V-BLIINDS [49] and VIIDEO [44] extend NSS to the temporal domain to address the BVQA task. TLVQM [26] integrates two-level features—spatial high-complexity and temporal low-complexity features—to measure complex spatial and temporal distortions. VIDEVAL [62] ensembles diverse handcraft features sourced from classical BI/VQA models to create a strong feature representation for VQA. In summary, knowledge-driven BVQA models typically incorporate temporal-related features into BIQA models or extend the 2D quality features into 3D, demonstrating the critical importance of temporal information in video quality evaluation. However, due to insufficient understanding of human perception in quality assessment, knowledge-driven BVQA models tend to underperform on in-the-wild videos.

Data-driven models have gained popularity with the advancement of DNNs, aiming to automatically learn quality-aware features from labeled visual data [4], [9], [11], [27], [28], [56], [65], [67], [68], [69], [73], [76], [86]. For example,

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Li et al. [28] use GRU [7] to capture the temporal relationship among the semantic features extracted by ResNet-50 [21]. Liu et al. [38] jointly optimize the feature extractor and the regressor for quality assessment and compression distortion classification. Wang et al. [65] utilizes the compression level, video content, and distortion type derived from independent networks to evaluate the quality of user-generated videos. Li et al. [27] combine the CNN-based spatial model and RNN-based temporal model to improve the VQA task. Zhu et al. [86] extract spatial distortion from frame feature statistics and local motion from frame feature difference, which are fed to a transformer network for the motion aware interaction learning. Wu et al. [68] proposed DisCoVQA, featuring a core spatiotemporal distortion extraction (STDE) module to extract multi-level spatiotemporal features. Sun et al. [56] propose SimpleVQA, an efficient and effective BVQA framework that uses sparse high-resolution key frames to extract spatial features and dense low-resolution frames to extract motion features. This framework is further streamlined to Minimalistic BVQA [58], which is used to analyze the characteristics of VQA datasets. Wu et al. [67] propose FAST-VQA, which samples video fragments (*i.e.*, the spatially spliced and temporally aligned patches) as input to a modified video Swin Transformer [39] to train an end-to-end BVQA model. They [69] later ensemble an aesthetics perception branch to FAST-VQA to better extract spatial features. Xing et al. [73] utilize a CLIP-based transformer to encode quality text descriptions and integrates the extracted features with spatiotemporal features to achieve enhanced video representations. De et al. [11] present an NR-IQA model by leveraging the benefits of latent diffusion models and achieve superior generalization in IQA task. Gao et al. [16] introduce a novel NR-IQA method that predicts the image quality score distribution to estimate the MOS. Although the methods achieve astonishing performance, they still suffer from compromised generalization abilities in VQA task.

B. Large Multimodal Models for Visual Quality Assessment

The advent of large multimodal models, typified by GPT-4V, has triggered keen interest in leveraging such techniques to quality assessment tasks [84]. Recent studies have explored the potential of adopting LMMs for visual quality assessment [3], [70], [71], [72]. For the IQA task, Zhu et al. [87] compare the performance of different LMMs employing the two-alternative forced choice (2AFC) prompting strategy. Wu et al. [70] investigate the use of LMMs to predict quantifiable quality scores by extracting the softmax pooling result from the logits of two frequent tokens: 'good' and 'poor'. However, for tasks that require a precise visual quality understanding, LMMs without instruction fine-tuning have shown inferior performance compared to traditional methods [70]. This is partly because LLMs struggle to produce accurate responses without a deep alignment between visual representation and scores. To address this, they proposed Q-instruct [71], a fine-tuned LMM model that leverages instructed question-answering datasets focused on low-level visual scoring. Chen et al. [4] fine-tunes a vision-language model to generate image quality descriptions for computed tomography, which were input to ChatGPT for quality scores. Further, Wu et al. [72] propose a human-like syllabus to teach LMMs using text-defined rating levels rather than generating scores directly.

For the VQA task, previous LMM methods only model intra-frame information through discrete frames, leading to a significant loss of temporal information. To address these issues, some works attempt to leverage ensemble methods to incorporate temporal features. For example, Wang et al. [9] combine Q-Align [72] and DOVER [69] to capture richer temporal features from DOVER. Their commendable performance in the AIS 2024 VQA Challenge [9] demonstrates that temporal information remains a critical factor in VQA tasks.

III. Methods

A. Problem Setting

For an LMM-based video quality assessment model, we define $\mathbf{X}$ as the video input, $\mathbf{T}$ as the text input, and $\hat{\mathbf{Q}}$ as the video quality, including quality scores or levels output. $\hat{\mathbf{Q}}$ is an approximation of the ground-truth video quality $\mathbf{Q}$. Then, we define the assessment model $\mathcal{F}$ as follows:

$$\hat{\mathbf{Q}} = \mathcal{F}(\mathbf{X}, \mathbf{T}). \tag{1}$$

B. Q&A Instruction Prompts

Large instruction-tuned multimodal models (*i.e.*, fine-tuned to respond to instructions) showcase an impressive ability for new tasks. Instruction prompt generation is a method for inducing instruction following capabilities with minimal human labeled data [47]. Its effectiveness has been validated through extensive instruction tuning experiments [66]. In our work, we construct Q&A instruction prompts using videos and their MOS, as illustrated in Fig. 2. The question part includes quality prompts and videos. Quality prompts can be divided into three parts: system prompts, instruction prompts, and response restrictions. System prompts delineate the role of the LMM in this task, ensuring that the LMM has a clear understanding of the task. Instruction prompts pose a direct question to the LMM, as a conversational distillation of the task description. Response restriction defines the desired format for the model's answers. A typical example of these prompts is shown in Fig. 2.

Specifically, we construct pair-wised Q&A instruction prompts based on the VQA benchmark datasets [22], [54], [64], [76], [78]. The basic analysis of these datasets is shown in Table III. Note that each video-score sample in VQA datasets results in two Q&A instruction prompts, separately for the quality score regression task and the quality classification task. For the quality score regression task, we take scores as the answer in Q&A pairs. For the classification task, we equally divide dataset samples into three levels—*'good'*, *'fair'*, and *'poor'*—based on their quality scores, and take these levels as the classification responses. To ensure the diversity of the training dataset, we generate 2,000 templates of instructions by GPT-4 such as: "Would you mind calculating the video quality score with the help of these frames?". Meanwhile, the visual tokens `<image-i>` extracted from video chunks and key frames are inserted followed by quality prompts during our training.

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TABLE III

DATASET STATISTICS OF FIVE EXISTING VQA DATASETS. WE ALSO SHOW THE DATA DESCRIPTION OF OUR PROPOSED INSTRUCTION Q&A PAIRS. FOR EACH VIDEO AND ITS QUALITY SCORE, WE CONSTRUCT TWO Q&A PAIRS FOR TRAINING

Dataset	Dataset Statistics				Instruction Q&A	
	# of videos	Resolution	Duration	Source	# of pairs	Annotator
LSVQ	3.8k	99p-4K	5-12s	Internet Archive & YFCC100M [61]	76k	GPT-4
KoNViD-1k	1.2k	540p	8s	YFCC100M [61]	2.4k	GPT-4
LIVE-VQC	0.6k	240p-1080p	10s	Camera-captured	1.2k	GPT-4
YouTube-UGC	1.5k	360p-4K	20s	YouTube	3k	GPT-4
LIVE-YT-Gaming	0.6k	360p-1080p	8-9s	Internet & software-recorded	1.2k	GPT-4

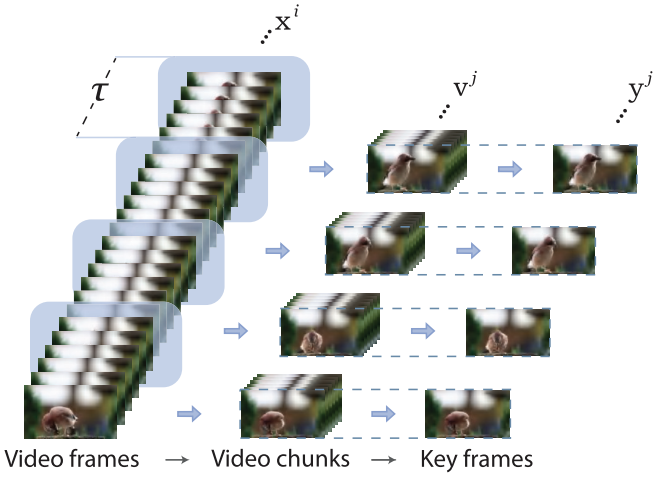

Fig. 3. **Illustration of the video preprocessor**. We slice the video into non-overlapping consecutive chunks every τ frames. The first frame of j-th chunk $\mathbf{v}^j$ is selected as key frame $\mathbf{y}^j$.

C. Video Preprocessing

To balance quality assessment efficacy and algorithmic complexity, we perform spatiotemporal downsampling operations during video preprocessing. Consider a video $\mathbf{X} = \{\mathbf{x}^i\}_{i=0}^{N-1}$, where $\mathbf{x}^i \in \mathbb{R}^{H\times W\times 3}$ represents the i-th frame with a resolution of $H\times W$, and N is the total number of frames. As illustrated in Fig. 3, we aim to extract video chunks $\mathbf{V} \in \mathbb{R}^{K\times\tau\times H\times W\times 3}$ for temporal feature extraction and key frames $\mathbf{Y} \in \mathbb{R}^{K\times H\times W\times 3}$ for spatial feature extraction. Thus, we slice the video in the temporal domain to derive non-overlapping consecutive video chunks $\mathbf{V} = \{\mathbf{v}^j\}_{j=0}^{K-1}$. Each video chunk has the same frame length τ and $K = \lfloor N/\tau \rfloor$. $\lfloor\cdot\rfloor$ is the floor function. So we define the j-th video chunk $\mathbf{v}^j$ as:

$$\mathbf{v}^j = \{\mathbf{x}^i\}_{i=\tau j}^{\tau(j+1)}, \tag{2}$$

Then, we select the first frame $\mathbf{v}^j$ as the j-th key frame $\mathbf{y}^j$:

$$\mathbf{y}^j = \mathbf{x}^{\tau j}, \tag{3}$$

Subsequently, video chunks V and key frames Y are processed by the vision encoder and projector to obtain visual tokens that serve as input to the LLM decoder.

D. Model Structure

Videos comprising sequences of frames depict motion scenes that differ from static images [39]. Therefore, the characteristics of both spatial and temporal dimensions are crucial for understanding videos [37]. Previous works have relied on discrete frames of videos to extract visual features, such as Video-ChatGPT [40]. While discrete frames provide a fundamental basis for understanding videos, they do not make use of the temporal features of continuous frames, leading to overlooked issues such as jitter, lagging, flickering, etc. Thus, to address spatial and temporal distortion issues in videos, we propose a novel spatiotemporal enhanced encoder—a two-branch encoder designed to enhance the capture of spatial and temporal information in videos.

To detail the soundness of our technical design, we compare our LMM-VQA with two similar approaches, LLaVA [35] and Video-LLaVA [32] respectively. As shown in Fig. 4, LLaVA only takes the images as input without other external modalities. Different from LLaVA, Video-LLaVA takes images and discrete video frames as input and uses different vision encoders for them. With a small modality gap between images and discrete video frames, Video-LLaVA performs well with the shared V-L translator in video understanding tasks. In our task, temporal features are quite different from pixel-level video representation. Thus, we propose a spatiotemporal encoder with each independent projector to facilitate the modality alignment.

Algorithm 1 LMM-VQA

Input: video chunks $\mathbf{V}$, key frames $\mathbf{Y}$, pair-wised instruction text prompts: $\mathbf{T}$

for $\mathbf{y}^j$ in $\{\mathbf{y}^0, \mathbf{y}^1, \cdots, \mathbf{y}^{K-1}\}$ **do**
  $\mathbf{F}_{\mathbf{sp}}^j = f_{sp}(\mathbf{y}^j)$
end for
All spatial vision embeddings: $\mathbf{F}_{\mathbf{sp}} \in \mathbb{R}^{N_p\times C_{sp}}$
for $\mathbf{v}^j$ in $\{\mathbf{v}^0, \mathbf{v}^1, \cdots, \mathbf{v}^{K-1}\}$ **do**
  $\mathbf{F}_{\mathbf{tp}}^j = f_{tp}(\mathbf{v}^j)$
end for
All temporal vision embeddings: $\mathbf{F}_{\mathbf{tp}} \in \mathbb{R}^{1\times C_{tp}}$
Alignment for spatial token:
$\mathbf{Z}_{\mathbf{sp}} = f_{ViT}\left(\mathbf{F}_{\mathbf{sp}}\right)$
Alignment for temporal token:
$\mathbf{Z}_{\mathbf{tp}} = Mean\left(f_{MLP}\left(\mathbf{F}_{\mathbf{tp}}\right)\right)$
Aggregated input token:
$\mathbf{Z}_{\mathbf{all}} = f_{cat}\left(\mathbf{Z}_{\mathbf{sp}}, \mathbf{Z}_{\mathbf{tp}}, \mathbf{Z}_{\mathbf{text}}\right)$
Generate quality score or level by LLM decoder:
$\mathbf{Z}_{\mathbf{quality}} = \mathcal{D}\left(\mathbf{z}_\ell \mid \mathbf{Z}_{\mathbf{all}}, \mathbf{z}_{<\ell}\right)$

Output: $\mathbf{Z}_{\mathbf{quality}}$

Algorithm 1 illustrates the pipeline of LMM-VQA, starting from key frames and video chunks. Specifically, as shown in, for the spatial branch, a 2D visual encoder (ViT-L/14 [48] in

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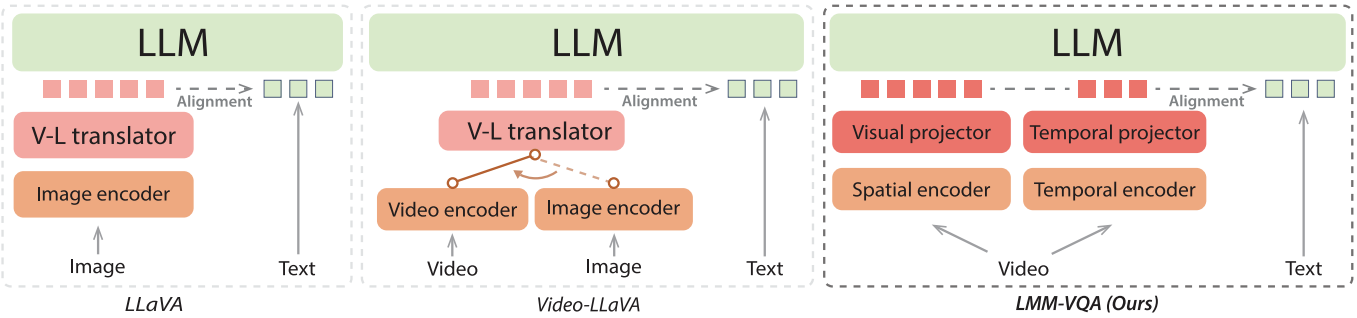

Fig. 4. **Technical comparisons with other methods**. **Left:** LLaVA takes the images as input only; **Middle:** the core architecture of Video-LLaVA. It shares a unified V-L translator for images and video; **Right:** our proposed method. We use two separate projectors for modality alignment to bridge the modality gap between video and text.

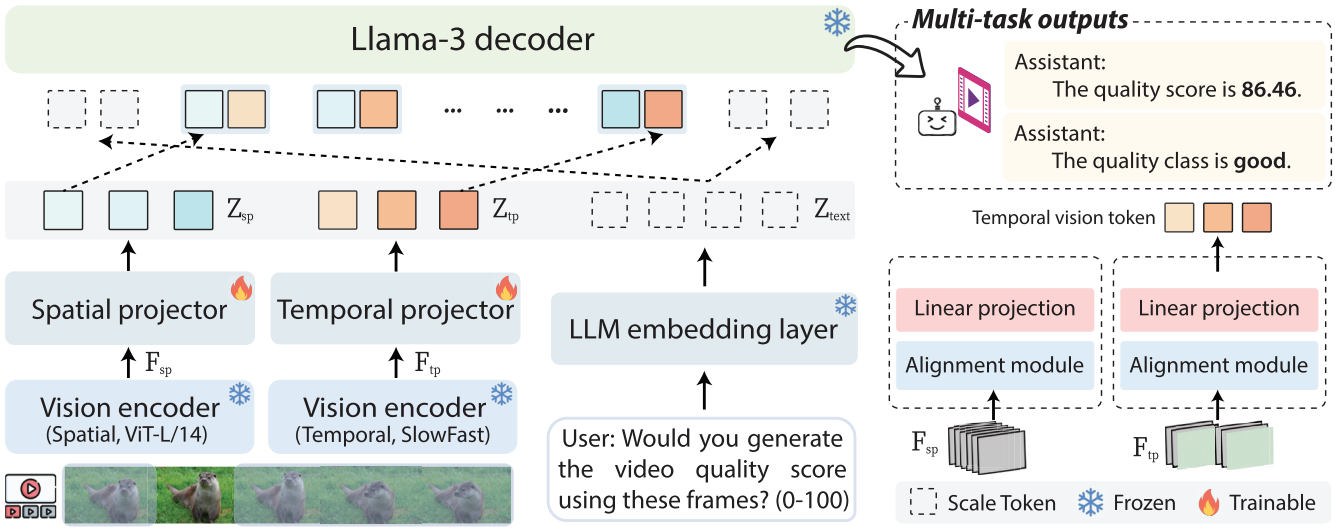

Fig. 5. **The framework of LMM-VQA**: LMM-VQA takes video frames as input, and generates responses of quality scores. The process initiates with two vision encoders that transform input frames into spatiotemporal visual features. These visual features are incorporated into two projection modules to generate visual tokens that are aligned with the text-related language space. Meanwhile, the text decoder produces scale tokens based on the quality prompt from users. Then, text-guided spatial and temporal tokens, and quality prompt tokens are aggregated as input for LLMs to generate the final answers.

Fig. 5) transforms the input key frames $\mathbf{Y}$ into spatial features $\mathbf{F_{sp}}$:

$$\mathbf{F_{sp}} = f_{sp}(\mathbf{Y}), \mathbf{F_{sp}} \in \mathbb{R}^{K \times N_p \times C_{sp}}, \tag{4}$$

where $N_p = H/p \times W/p$ indicate the number of image patches and and C_{sp} is the number of spatial embedding dimensions, respectively. For the temporal branch, a 3D visual encoder (SlowFast in Fig. 5) extracts temporal features $\mathbf{F_{tp}}$:

$$\mathbf{F_{tp}} = f_{tp}(\mathbf{V}), \mathbf{F_{tp}} \in \mathbb{R}^{K \times 1 \times C_{tp}}, \tag{5}$$

where C_{tp} is the number of temporal embedding dimensions.

To address the disparity between vision and text modalities, we design a spatiotemporal projector to map visual tokens into language space for modality alignment, as illustrated in Fig. 5. Specifically, we introduce a **spatial projector** which comprises a single-layer vision transformer (ViT) and a one-layer MLP, to map spatial features into the language space. Following this, we use a one-layer MLP as a **temporal projector** to align temporal features with the language space. We formulate these processes as:

$$\mathbf{Z_{sp}} = f_{ViT}\left(\mathbf{F_{sp}}\right), \tag{6}$$

$$\mathbf{Z_{tp}} = Mean\left(f_{MLP}\left(\mathbf{F_{tp}}\right)\right). \tag{7}$$

A text decoder generates scalable text tokens $\mathbf{F_{text}}$ from quality prompts $\mathbf{T}$ (*i.e.*, LLM embedding layer in Fig. 5). Spatial tokens $\mathbf{Z_{sp}}$, temporal tokens $\mathbf{Z_{tp}}$ and text tokens $\mathbf{Z_{text}}$ are concatenated in the context length dimension f_{cat}, defined as $\mathbf{Z_{all}}$:

$$\mathbf{Z_{all}} = f_{cat}\left(\mathbf{Z_{sp}}, \mathbf{Z_{tp}}, \mathbf{Z_{text}}\right). \tag{8}$$

These text-guided visual tokens and quality prompt tokens are aggregated to generate the final output tokens, which represent the predicted quality scores or levels.

We utilize the Llama-3 model [13] as the language decoder and output the predicted answers in an auto-regressive manner:

$$\mathbf{Z_{quality}} = \mathcal{D}\left(\mathbf{z}_\ell \mid \mathbf{Z_{all}}, \mathbf{z}_{<\ell}\right), \tag{9}$$

where $\mathbf{Z_{quality}} = \{\mathbf{z}_l\}_{l=0}^{L-1}$ is the output text sequence follows the input tokens $\mathbf{Z_{all}}$ and $\mathcal{D}(\cdot)$ is the Llama-3 decoder. Maximum token length is set as L. $\mathbf{z}_l \in \{0,1\}^{|\mathbb{S}|}$ and $\mathbb{S}$ denote the vocabulary set. With well-trained LMM-VQA, $\mathbf{Z_{quality}}$ represents an approximation $\hat{\mathbf{Q}}$ to the true perceptual quality $\mathbf{Q}$. The training process uses a cross-entropy loss:

$$\mathcal{L} = -\sum_{\ell=0}^{L-1} \log \boldsymbol{p}\left(\mathbf{z}_\ell \mid \mathbf{Z_{all}}, \mathbf{z}_{<\ell}\right). \tag{10}$$

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We use pair-wised Q&A instruction datasets for training our model on the quality score regression task, referred to as LMM-VQA (wo/multi-task). Additionally, we also leverage Q&A instruction datasets for both quality score regression and quality classification tasks to train another version of the model, termed as LMM-VQA (w/multi-task).

IV. Experiment

A. Experimental Settings

1) Implementation Details: We utilize *Llama-3-8b-Instruct* [13] as the decoder, *clip-vit-large-L/14* [48] and *SlowFast* [15] as the spatial and temporal encoders, respectively. We utilize paired Q&A instruction prompts as our training data and train our model in an end-to-end manner for VQA tasks. During the training phase, the parameters of both the decoder and the vision encoders are frozen. We only train the spatial and temporal projectors to align modalities between visual and text tokens. Training experiments are conducted on eight NVIDIA A800 GPUs, while inference is performed on two NVIDIA A6000 GPUs. The batch size is 32, with a learning rate of 0.001. The warmup steps are 10% of the training data size, and training epochs are set as 6 to achieve convergence. The temperature coefficient is set to 1 and beam search is not employed. The maximum length of the input and output tokens is 4096 as the stopping condition. The video input resolution $H \times W$ is 224×224 and the patch size p is typically set to 14 for the ViT-L/14. The length of video chunk τ is configured to equal the video frame rate, *i.e.*, one video chunk and one key frame are sampled per second. The length of the spatial and temporal tokens are 257 and 1, respectively. The length of the text token varies depending on the size of the quality prompt.

2) Evaluation Criteria: We select Spearman Rank-Order Correlation Coefficient (SRCC) and Pearson Linear Correlation Coefficient (PLCC) as metrics for performance evaluation, which indicate the prediction monotonicity and prediction accuracy. It should be noted that SRCC and PLCC range from 0 to 1.0, where larger values indicate better results.

3) VQA Datasets: For instruction tuning and evaluation, we construct our Q&A instruction prompts based on five datasets: LSVQ [76], KoNViD-1k [22], YouTube-UGC [64], LIVE-VQC [54], LIVE-YT-Gaming [78], with details provided in Table III. LSVQ, the largest VQA dataset in our study, comprises 38,811 videos, covering a wide range of resolutions and frame rates. KoNViD-1K consists of 1,200 videos of fixed resolution sourced from the Internet, primarily focusing on spatial distortions. LIVE-VQC includes 585 videos captured directly by various mobile cameras, showcasing a broad range of natural scenes, different lighting conditions, and diverse motion levels. YouTube-UGC consists of 1,500 YouTube video clips, each 20 seconds in duration, with resolutions spanning from 360p to 4K. Targeting streamed gaming videos, LIVE-YT-Gaming contains 600 gaming videos belonging to 59 games.

For the LSVQ dataset, we follow the original dataset splitting [76], in which the training set consists of 28,013 videos, while 7,220 and 3,561 videos for the default test and test-1080p sets respectively. Taking the LSVQ dataset as the training set, we randomly sample 10% videos from the training set for validation and report the results on the two intro test sets. For the remaining four smaller datasets, we use 5-fold cross-validation in fine-tuning experiments and report the results on the entire dataset for OOD evaluation.

4) Competing Methods: We choose several representative VQA methods for comparison. The competing methods can be roughly classified into three groups: 1) three handcrafted models: TLVQM [26], VIDEVAL [62], and RAPIQUE [63]; 2) seven DNN-based methods: VSFA [28], PVQ [76], BVQA [27], FAST-VQA [67], SimpleVQA [56], MinimalisticVQA [58], and DOVER [69]; 3) an LMM-based method: Q-Align [72].

B. Performance Comparison

1) Quality Scoring Task: To comprehensively evaluate our proposed method, we assess our model with an in-sample, OOD, and fine-tuning manner.

Table IV presents the in-sample results for BVQA models on the LSVQ and LSVQ-1080p datasets. All models are trained on the LSVQ training set ($\text{LSVQ}_{\text{train}}$) and evaluated on $\text{LSVQ}_{\text{test}}$ and $\text{LSVQ}_{1080\text{p}}$ test sets. It is observed that LMM-VQA consistently outperforms all competing methods in terms of both SRCC and PLCC. Specifically, for $\text{LSVQ}_{\text{test}}$, LMM-VQA (wo/multi-task) achieves an SRCC of 0.913 and a PLCC of 0.914, outperforming SOTA models such as DOVER [69] and Q-Align [72]. Further, LMM-VQA (w/multi-task) performs better than LMM-VQA (wo/multi-task), which indicates that multi-task learning for quality score regression and quality level classification enhances the quality understanding of videos for LMMs, then improves the prediction performance of quality scores. On the $\text{LSVQ}_{1080\text{p}}$ set, LMM-VQA also leads with an SRCC of 0.879 and a PLCC of 0.898, significantly surpassing DNN-based models like DOVER [69], and LMM-based method like Q-Align [72]. These results show that our proposed LMM-VQA achieves the best performance in both SRCC and PLCC across different test sets, demonstrating its effectiveness in VQA tasks.

We conduct OOD evaluations to further evaluate the generalization capability of our LMM-VQA. Table V shows that LMM-VQA (wo/multi-task) significantly outperforms the other models on KoNViD-1K [22], YouTube-UGC [64], LIVE-VQC [64] and LIVE-YT-Gaming [78] datasets. For KoNViD-1k, LMM-VQA (wo/multi-task) achieves an SRCC of 0.875 and a PLCC of 0.876, outperforming other LMM-based models such as Q-Align (0.865 and 0.877), along with MinimalisticVQA (0.862 and 0.863), the best DNN-based models. For YouTube-UGC and LIVE-VQC datasets, which present significant challenges due to poor exposure conditions and a variety of motion types, our proposed LMM-VQA (wo/multi-task) achieves an SRCC/PLCC of 0.858/0.877 on the YouTube-UGC dataset and 0.831/0.863 on the LIVE-VQC dataset. This superior performance can largely be attributed to the generalization capabilities of LLMs, which are pre-trained on the vast unlabeled text corpus. This extensive pre-training enables LMM-VQA to effectively understand the nuances of MOS, adapt to diverse prompts in VQA, and generalize

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TABLE IV

IN-SAMPLE PERFORMANCE OF LMM-VQA AGAINST COMPETITIVE METHODS. ALL METHODS ARE TRAINED ON $\text{LSVQ}_{\text{TRAIN}}$ AND EVALUATED ON TWO INTRA-DATASET ($\text{LSVQ}_{\text{TEST}}$ AND $\text{LSVQ}_{1080\text{P}}$). W/WO REFERS TO WHETHER OUR MODEL IS TRAINED ON MULTI-TASKS OR NOT. PAQ-2-PIQ [77] IS AN IQA MODEL THAT PRE-TRAINED RESNET-18 [21] ON THE LIVE-FB DATASET [77]. UNIQUE [83] IS AN IQA MODEL THAT PRE-TRAINED RESNET-50 [21] ON FOUR FUSED IQA DATASETS [8], [14], [17], [23]. OUR METHODS ARE MARKED IN GRAY AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Training Set: **$\text{LSVQ}_{\text{train}}$**	*→Testing Set:*	**$\text{LSVQ}_{\text{test}}$**		**$\text{LSVQ}_{1080\text{p}}$**	
Method	**Feature Extraction Modules**	SRCC	PLCC	SRCC	PLCC
TLVQM [26] (TIP 2019)	Handcrafted features	0.772	0.774	0.589	0.616
VIDEVAL [62] (TIP 2021)	Handcrafted features	0.794	0.783	0.545	0.554
VSFA [28] (ACMMM 2019)	ResNet-50 based	0.801	0.796	0.675	0.704
PVQ [76] (CVPR 2021)	PaQ-2-PiQ & 3D ResNet-18 based	0.827	0.828	0.711	0.739
BVQA [27] (TCSVT 2022)	UNIQUE & SlowFast based	0.852	0.854	0.772	0.788
FAST-VQA [67] (ECCV 2022)	Video Swin-T	0.876	0.877	0.779	0.814
SimpleVQA [56] (ACMMM 2022)	ResNet-50 & SlowFast based	0.867	0.861	0.764	0.803
MinimalisticVQA [58] (TPAMI, 2023)	Swin-B & SlowFast based	0.881	0.879	0.781	0.820
DOVER [69] (ICCV 2023)	ConvNeXt-T & Video Swin-T based	0.886	0.887	0.795	0.830
Q-Align [72] (ICML, 2024)	mPLUG-Owl-2 based	0.883	0.882	0.797	0.830
LMM-VQA (wo/multi-task)	ViT & SlowFast & Llama-3 based	0.913	0.914	0.879	0.898
LMM-VQA (w/multi-task)	ViT & SlowFast & Llama-3 based	**0.916**	**0.919**	**0.891**	**0.899**

TABLE V

OOD PERFORMANCE OF LMM-VQA AGAINST COMPETITIVE METHODS. ALL METHODS ARE TRAINED ON $\text{LSVQ}_{\text{TRAIN}}$ AND EVALUATED ON FOUR OOD DATASETS INCLUDING KONVID-1K, YOUTUBE-UGC, LIVE-VQC, AND LIVE-YT-GAMING. NA IN TABLE REFERS TO THAT THE RESULTS CANNOT BE PRODUCED

Training Set: **$\text{LSVQ}_{\text{train}}$** *→Testing Set:*	**KoNViD-1k**		**YouTube-UGC**		**LIVE-VQC**		**LIVE-YT-Gaming**	
Method	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
TLVQM [26] (TIP 2019)	0.732	0.724	0.685	0.692	0.670	0.691	0.740	0.775
VSFA [28] (ACMMM 2019)	0.784	0.794	0.733	0.728	0.753	0.795	0.669	0.698
VIDEVAL [62] (TIP 2021)	0.751	0.741	0.687	0.709	0.630	0.640	NA	NA
PVQ [76] (CVPR 2021)	0.791	0.795	0.742	0.754	0.770	0.807	NA	NA
BVQA [27] (TCSVT 2022)	0.839	0.830	0.802	0.792	0.816	0.824	0.690	0.740
SimpleVQA [56] (ACM MM 2022)	0.839	0.841	0.811	0.813	0.730	0.780	0.666	0.734
FAST-VQA [67] (ECCV 2022)	0.859	0.855	0.730	0.747	0.823	0.844	0.620	0.666
MinimalisticVQA [58] (TPAMI 2024)	0.862	0.863	0.826	0.833	0.798	0.837	0.711	0.755
Q-Align [72] (ICML, 2024)	0.865	0.877	0.834	0.848	0.773	0.830	0.611	0.687
LMM-VQA (wo/multi-task)	0.875	0.876	**0.858**	**0.877**	**0.831**	**0.863**	0.783	0.789
LMM-VQA (w/multi-task)	**0.901**	**0.902**	0.748	0.761	0.767	0.805	**0.816**	**0.801**

TABLE VI

THE FINE-TUNING PERFORMANCE OF LMM-VQA AGAINST COMPETITIVE METHODS. ALL METHODS ARE TRAINED ON THE $\text{LSVQ}_{\text{TRAIN}}$ AND FINE-TUNED ON FOUR OOD DATASETS INCLUDING KONVID-1K, YOUTUBE-UGC, LIVE-VQC, AND LIVE-YT-GAMING. WE PERFORMED A FIVE-FOLD CROSS-VALIDATION TO EVALUATE PERFORMANCE. THE RESULTS PRESENTED IN THE TABLE ARE THE AVERAGE VALUES DERIVED FROM THESE FIVE EXPERIMENTS

Fine-tuning & Testing Set:	**KoNViD-1k**		**YouTube-UGC**		**LIVE-VQC**		**LIVE-YT-Gaming**	
Method	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
VSFA [28] (ACMMM 2019)	0.794	0.799	0.787	0.789	0.718	0.771	0.784	0.819
RAPIQUE [63] (OJSP 2021)	0.811	0.819	0.774	0.781	0.740	0.764	0.771	0.815
BVQA [27] (TCSVT 2022)	0.834	0.836	0.818	0.826	0.834	0.842	NA	NA
SimpleVQA [56] (ACMMM 2022)	0.856	0.860	0.847	0.856	0.845	0.859	0.861	0.866
FAST-VQA [67] (ECCV 2022)	0.891	0.892	0.852	0.855	0.849	0.865	0.869	0.880
DOVER [69] (ICCV 2023)	0.909	0.906	0.890	0.891	0.860	0.875	0.852	0.868
MinimalisticVQA [58] (TPAMI 2024)	0.889	0.890	0.890	0.891	0.842	0.854	0.857	0.888
LMM-VQA (wo/multi-task)	**0.929**	**0.930**	**0.901**	**0.897**	**0.891**	**0.903**	**0.914**	**0.901**

across various datasets. For the LIVE-YT-Gaming dataset that focuses on a specific type of video, LMM-VQA (wo/multi-task) has also achieved SOTA performance, demonstrating its excellent generalization capability. However, LMM-VQA (w/multi-task) exhibits substantial performance fluctuations. It achieves exceptionally excellent results on the KoNViD-1K and LIVE-YT-Gaming datasets but falls short of SOTA performance on the YouTube-UGC and LIVE-VQC datasets.

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TABLE VII

THE CLASSIFICATION PERFORMANCE OF LMM-VQA AGAINST EXISTING SOTA METHODS. ALL METHODS ARE TRAINED ON $\text{LSVQ}_{\text{TRAIN}}$ AND EVALUATED ON TWO IN-SAMPLE DATASETS ($\text{LSVQ}_{\text{TEST}}$ AND $\text{LSVQ}_{1080\text{P}}$) AND FOUR OOD DATASETS (KONVID-1K, YOUTUBE-UGC, LIVE-VQC, AND LIVE-YT-GAMING)

Training Set: **$\text{LSVQ}_{\text{train}}$** *→Testing Set:*	**$\text{LSVQ}_{\text{test}}$**				**$\text{LSVQ}_{1080\text{p}}$**				**KoNViD-1k**			
Level	Poor	Fair	Good	Total	Poor	Fair	Good	Total	Poor	Fair	Good	Total
Q-Align [72] (ICML, 2024)	0.880	0.288	**0.931**	0.702	0.558	0.142	0.973	0.561	0.922	0.415	0.797	0.709
MinimalisticVQA [58] (TPAMI 2024)	0.797	0.677	0.756	0.744	0.711	0.668	0.614	0.664	0.841	0.716	0.248	0.602
LMM-VQA (w/multi-task)	**0.914**	**0.843**	0.925	**0.950**	**0.953**	**0.897**	**0.977**	**0.955**	**0.940**	**0.943**	**0.967**	**0.950**
Testing Set:	**YouTube-UGC**				**LIVE-VQC**				**LIVE-YT-Gaming**			
Level	Poor	Fair	Good	Total	Poor	Fair	Good	Total	Poor	Fair	Good	Total
Q-Align [72] (ICML, 2024)	**0.976**	0.294	0.034	0.431	0.451	0.788	0.467	0.568	0.389	**0.914**	0.029	0.440
MinimalisticVQA [58] (TPAMI 2024)	0.923	0.395	0.544	0.620	0.653	0.347	**0.915**	0.641	0.390	0.626	0.549	0.522
LMM-VQA (w/multi-task)	0.960	**0.830**	**0.727**	**0.838**	**0.933**	**0.839**	0.854	**0.875**	**0.667**	0.514	**0.684**	**0.625**

The reason is that the quality distribution of videos differs between the training set LSVQ and two test sets, YouTube-UGC and LIVE-VQC are different, thus making the quality levels of the same video vary across these sets. So, multi-task training with quality level classification may lead to LMM-VQA misunderstanding the quality levels of videos in YouTube-UGC and LIVE-VQC, thereby affecting their quality scores.

Table VI illustrates the fine-tuning performance against different methods. From the results, we can see that LMM-VQA (wo/multi-task) maintains SOTA performance across all benchmark datasets. For the LIVE-VQC dataset, our LMM-VQA yields an SRCC and PLCC of 0.891 and 0.903, demonstrating improvements of 7.2% and 4.6% compared with the model without fine-tuning, respectively. For the LIVE-YT-Gaming dataset, LMM-VQA still attains an SRCC and PLCC of 0.914 and 0.901, showing improvements of 12%, 12.5% compared to the model without fine-tuning. Further, compared to the previous SOTA method, MinimalisticVQA, we obtain an average improvement of 4.6% SRCC and 3.1% PLCC in performance. These results show that LMM-VQA can be successfully trained on small-scale VQA datasets without overfitting.

2) Quality Classification Task: As we introduce quality level classification as an auxiliary task for multi-task training, we evaluate the performance of the proposed model in predicting quality levels. We calculate the prediction accuracy across four scenarios: 'Poor', 'Fair', 'Good', and 'Total', where 'Total' refers to accuracy calculated across all test quality levels. All models are trained on the same dataset ($\text{LSVQ}_{\text{train}}$) and evaluated on datasets including $\text{LSVQ}_{\text{test}}$, $\text{LSVQ}_{1080\text{p}}$, KoNViD-1k, YouTube-UGC, LIVE-VQC, and LIVE-YT-Gaming datasets. Table VII shows that LMM-VQA significantly outperforms the traditional DNN-based model like MinimalisticVQA and LMM-based model like Q-Align across several datasets.

C. Ablation Study

1) Temporal Tokens: Researchers [26], [27], [58] have demonstrated that the temporal relationship between frames of the video is very important for VQA. We model the temporal relationship to enhance the LMM's capabilities of capturing the temporal distortion cues in videos. Thus, to demonstrate the rationality of LMM-VQA, we discuss the effect of temporal tokens in our ablation study. Table VIII (I) and (II) compare the performance of our proposed LMM-VQA on two conditions: with temporal tokens (I) and without temporal tokens (II). We compare their performance on both the LSVQ test and test-1080p sets. From the results, we observe that the incorporation of temporal tokens consistently enhances the performance metrics across both datasets. For $\text{LSVQ}_{\text{test}}$, the metrics PLCC/SRCC rise from 0.855/0.887 to 0.889/0.891, respectively. Moreover, by integrating temporal tokens into Llama-3, we achieve improvements of 2.3% (PLCC) and 3.0% (SRCC) in the prediction accuracy of $\text{LSVQ}_{1080\text{p}}$.

TABLE VIII

ABLATION STUDY ON MOTION TOKENS AND MODEL STRUCTURE OF PROPOSED METHODS. BOTH SRCC AND PLCC METRICS ARE CALCULATED ON LSVQ TEST SETS. ALL MICRO-DESIGNS FURTHER IMPROVE THE PERFORMANCE OF THE PROPOSED METHOD

Micro-design	**$\text{LSVQ}_{\text{test}}$**		**$\text{LSVQ}_{1080\text{p}}$**	
	SRCC	PLCC	SRCC	PLCC
(I): *Llama-2 & ViT-L/14*	0.855	0.887	0.833	0.864
(II): *(I) with temporal tokens*	0.889	0.891	0.858	0.884
Improvement over (I)	+3.9%	+0.4%	+3.0%	+2.3%
(III): *(II) with Llama-3 as decoder*	0.913	0.914	0.879	0.898
Improvement over (I)	+6.7%	+3.0%	+5.5%	+3.9%
Improvement over (II)	+2.6%	+2.6%	+2.4%	+1.6%
(IV): *(III) with multi-task training*	0.916	0.919	0.891	0.899
Improvement over (I)	+7.1%	+3.6%	+7.0%	+4.1%
Improvement over (III)	+0.3%	+0.5%	+1.4%	+0.1%

We also visualize several predictions of LMM-VQA in Fig. 6. We use TVL1 algorithm [60] to compute optical flow values. These values are subsequently scalarized and spatially averaged to derive motion values between frames. Specifically, in case (a), the video exhibits stutter, a condition that has been discussed above and poses difficulties for several existing methods such as [72]. Under this condition, the model without temporal tokens significantly overestimates the scores of videos. By introducing temporal tokens, the accuracy of LMM-VQA is substantially improved. In case (b), for videos containing complex textures, the model that

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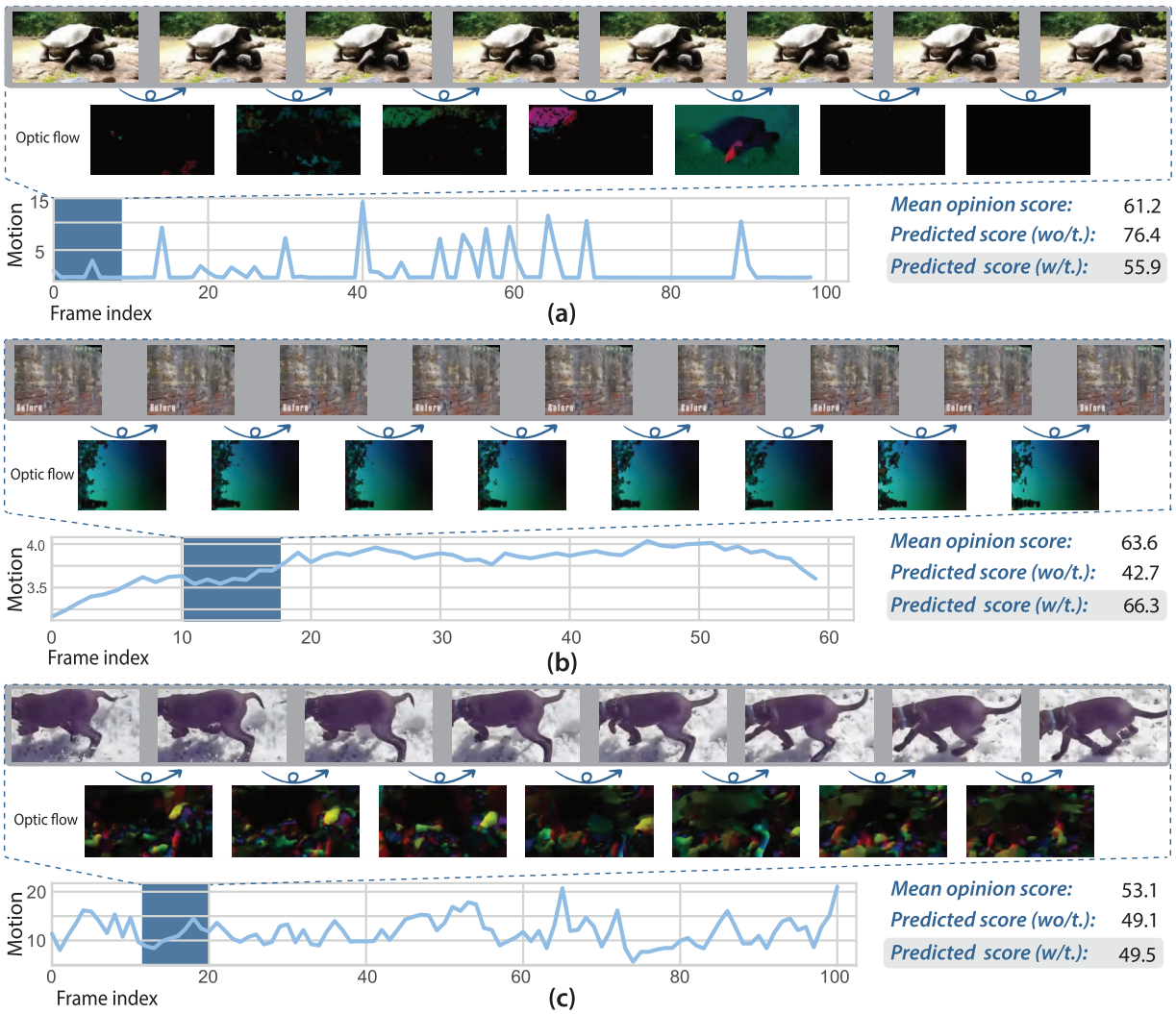

Fig. 6. **Visualization of predictions of the proposed model on LSVQ datasets**. We discuss three scenarios to discuss the effect of temporal tokens for VQA. (a) For lagging videos, introducing temporal tokens avoids the overestimation of quality scores.; (b) For complex textures of videos, motion tokens prevent the underestimation of quality scores. (c) In some smooth videos, adding motion tokens shows no improvement in prediction accuracy.

TABLE IX

ABLATION STUDY OF PROPOSED METHODS ON LSVQ TEST SETS. (A) ABLATION STUDY OF THE SPATIAL PROJECTOR. WE SELECT MLP, PERCEIVER, AND VIT MODULES FOR COMPARISON; (B) ABLATION STUDY ON TRAINING DATA PROPORTIONS. WE DESIGN FOUR SCENARIOS WITH VARYING FRACTIONS OF TRAINING DATA FOR EVALUATION. FOR EXAMPLE, WE RANDOMLY ALLOCATE A 25% SUBSET OF TRAINING DATASETS, AND TEST ON LSVQ DEFAULT TEST SET; (C) ABLATION STUDY ON THE LENGTH OF TEMPORAL TOKENS, TEST ON LSVQ DEFAULT TEST SET. RT REFERS TO THE RUNNING TIME ON A SPECIFIC VIDEO. DEFAULT SETTINGS ARE MARKED IN GRAY

	$LSVQ_{test}$		$LSVQ_{1080p}$	
	SRCC	PLCC	SRCC	PLCC
MLP	0.904	0.904	0.866	0.892
Perceiver	0.892	0.891	0.844	0.878
ViT	0.913	0.914	0.879	0.898

(a) Spatial projector.

	SRCC	PLCC
25%	0.758	0.801
50%	0.856	0.876
75%	0.878	0.881
100%	0.913	0.914

(b) Proportions of training data.

	SRCC	PLCC	RT/s
4	0.879	0.879	1.98
16	0.886	0.887	2.89
64	0.904	0.904	3.26
256	0.913	0.914	3.99

(c) The length of temporal tokens.

lacks temporal tokens is significantly lower than the labeled MOS. This discrepancy arises because, in the absence of temporal information, video frames with limited resolution may cause the model to mistake complex textures for noise. Such misinterpretations commonly occur in scenes featuring elements like rainy weather, grasslands, and textured wall surfaces. By introducing temporal tokens, the model significantly improves its prediction accuracy. For case (c), it is a normal smooth video, and the results indicate that adding temporal information does not improve the accuracy of predictions.

2) The Length of Temporal Tokens: Next, we conduct experiments with the length of temporal tokens for further investigation. With the temporal features of SlowFast [15], we

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TABLE X
QUANTITATIVE RESULTS ON VCGBENCH.DEFAULT SETTINGS ARE MARKED IN GRAY

Method	Using Subtitles	Information Correctness	Detailed Orientation	Contextual Understanding	Temporal Understanding	Consistency
LLaMA Adapter [82]	✗	2.03	2.32	2.30	1.98	2.15
Video LLaMA [80]	✗	1.96	2.18	2.16	1.82	1.79
Video Chat [30]	✗	2.23	2.50	2.53	1.94	2.24
Video-ChatGPT [40]	✗	2.40	2.52	2.62	1.98	2.37
BT-Adapter-7B [37]	✗	2.68	2.69	3.27	2.34	2.46
LLaMA-VID-7B [31]	✗	2.96	3.00	**3.53**	2.46	2.51
LMM-VQA	✗	**3.03**	**3.07**	**3.53**	**2.83**	**2.53**

set up $N_t = 256$ as the token length of the slow feature. We select the default test set of LSVQ for verification. Table IX lists several token lengths of the slow feature. The results clearly show that performance consistently improves across SRCC/PLCC metrics as N_t increases. Specifically, when N_t is increased to 64, we observe an obvious improvement of approximately 3% compared to $N_t = 4$. The model achieves optimal performance in SRCC and PLCC metrics with $N_t = 256$ per frame of video. However, this configuration results in an increased running time of 3.99 seconds per video, which is impractical. Thus, we select $N_t = 64$ as a trade-off, identifying it as the optimal number of temporal tokens.

3) Decoders: Next, we will discuss the effect of decoders. We select Llama-2-7b and Llama-3-7b for comparison. Table VIII (II) and (III) present the performance of two foundational models on two datasets: $\text{LSVQ}_{\text{test}}$ and $\text{LSVQ}_{1080\text{p}}$. The results indicate that the LMM-VQA with Llama-3 outperforms the Llama-2-based model across both datasets. Specifically, the performance metrics (PLCC/SRCC) for Llama-3 on $\text{LSVQ}_{\text{test}}$ increase from 0.889/0.891 to 0.913/0.914, and on the $\text{LSVQ}_{1080\text{p}}$ from 0.858/0.884 to 0.879/0.898.

4) The Task of Quality Classification: Table VIII (III) and (IV) compare the performance of our proposed LMM-VQA on two conditions: with and without multi-task training strategies respectively. The results illustrate that introducing classification tasks for training gains 7.1% SRCC and 3.6% benefits compared to baseline methods, suggesting that the multi-task training strategy can significantly improve model performance.

5) Proportions of Training Data: Recent work [25] has found power-law scalings relationships between performance and dataset size in LLMs. In this study, we examine the performance of our model across various proportions of training data to determine whether the current amount of training data is sufficient. Table IX illustrates the performance of LMM-VQA across different proportions of training data. The results show a clear increasing trend indicating that increasing the proportion of training data generally enhances the model's performance for VQA. Specifically, performance steadily improves from 0.758 at 25% of training data to 0.913 at 100% of training data. We can infer that the current volume of training data is insufficient. Collecting additional data for training may yield greater benefits from LMM-based VQA methods.

6) Modality Projection: Aligning representations from external modalities into language space remains a challenging task. In our proposed LMM-VQA, we integrate spatial and temporal tokens into the language space using two projectors, as shown in Figure 5. We use a ViT-based model for the spatial projector and an MLP layer for the temporal projector. Here, we discuss the impact of different spatial projectors, keeping the temporal projector unchanged. We compare three typical methods for modality projection, including Perceiver [24], MLP, and our proposed ViT-based module. As depicted in Table IX, our proposed ViT module achieves significant gains. Compared with Perceiver [24], our proposed ViT projector can be further enhanced and attains peak performance in all predicted metrics. For $\text{LSVQ}_{1080\text{p}}$, our proposed ViT projector stands out with SRCC/PLCC of 0.879/0.898, significantly surpassing MLP (0.866/0.892) and Perceiver (0.844/0.878). These findings suggest that the ViT projection layer is more effective in capturing and representing the relevant features, leading to superior performance.

D. Performance Evaluation on General Benchmark

To verify that our proposed model can also benefit general video understanding tasks, we assess its performance on the general VCGBench [40], detailed in Table X.

1) Training Pipeline: For the general benchmark, we propose a special version of LMM-VQA, which is trained with two stages, *i.e.*, modality alignment, and instruction tuning. For modality alignment, we train a spatial projector and a temporal projector, which projects the spatial and temporal features to the LLM's language space. We leverage a combined image captioning dataset that includes images from LAION [51], Conceptual Captions [53], and SBU [46] to align the visual feature with LLM's input space. In this stage, we primarily optimize the projectors in Fig. 5, while freezing the pre-trained modules like the LLM and vision encoders. In the stage of instruction tuning, we construct pair-wised instruction datasets to enhance the multimodal understanding of LLMs. In particular, we take multiple frames as input and design the predefined prompts in the following template: `Human: <prompt><image-1><image-2>..<image-N>, Assistant: <answer>`. The detailed contents of `<prompt>` and `<answer>` can be found in [40]. Further, we represent each image with 64 spatial tokens and 1 temporal token for

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Llama-3. In instruction tuning, all the modules are optimized except the frozen vision encoder.

2) Results Analysis: The VCGBench encompasses five critical aspects: information correctness, detail orientation, contextual understanding, temporal understanding, and consistency. From the results, we can see our model achieves SOTA performance in temporal understanding dimensions, which verified that our model can utilize the temporal tokens to improve video understanding. Further, the reason why our model can facilitate video understanding is that we encode both spatial and temporal features in SlowFast [15] into temporal tokens, so that those tokens contain the high-level temporal representation of videos. Then, these tokens introduce additional all-frame visual representations that are not captured by vision encoders with fixed frames. In the other evaluation dimensions, our model is comparable with the previous methods without subtitles.

V. Conclusion and Future Discussions

In this paper, we present LMM-VQA, an LMM-based VQA model that aims at better modeling spatial and temporal relationships to predict accurate scores and levels for VQA. LMM-VQA takes video frames as input, and generates the responses of quality scores with the quality levels. The process initiates with two vision encoders that separately transform input frames into spatial and temporal features. To better capture temporal distortions and align different modalities, we design two spatial and temporal projectors to integrate visual tokens into the language space for modality alignment. We utilize Llama-3 as our language decoder, the aligned visual tokens and quality prompt tokens are aggregated as input for Llama-3 to generate the final answers. The proposed LMM-VQA has reached SOTA performance on five popular VQA datasets and achieved excellent generalization ability among them.

However, unintended risks may still exist when LMM-VQA is unable to adequately analyze observations due to a limited understanding of quality regression, potentially leading to invalid outcomes in the VQA task. To address this issue, we may need to focus on improving the quality and diversity of the multimodal training data and refining the LMM's understanding of the relationship between video and quality score concepts. On the other hand, we can incorporate human feedback into LMM-VQA so that it can optimize the prediction performance of the quality scores iteratively. In the future, we will also discuss the reasoning process of LMM-VQA's generation to solve more complex problems including the better coverage of non-natural contents, and dealing with ambiguous quality scores.

References

[1] OpenAI et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[2] C. G. Bampis, Z. Li, and A. C. Bovik, "Spatiotemporal feature integration and model fusion for full reference video quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 8, pp. 2256–2270, Aug. 2019.

[3] Y. Chen, L. Liu, and C. Ding, "X-IQE: EXplainable image quality evaluation for text-to-image generation with visual large language models," 2023, *arXiv:2305.10843*.

[4] Z. Chen et al., "IQAGPT: Computed tomography image quality assessment with vision-language and ChatGPT models," *Vis. Comput. Ind., Biomed., Art*, vol. 7, no. 1, p. 20, Aug. 2024.

[5] Z. Chen and D. Wu, "Prediction of transmission distortion for wireless video communication: Analysis," *IEEE Trans. Image Process.*, vol. 21, no. 3, pp. 1123–1137, Mar. 2012.

[6] Z. Cheng et al., "VideoLLaMA 2: Advancing spatial–temporal modeling and audio understanding in video-LLMs," 2024, *arXiv:2406.07476*.

[7] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," 2014, *arXiv:1406.1078*.

[8] A. Ciancio, A. L. N. T. da Costa, E. A. B. da Silva, A. Said, R. Samadani, and P. Obrador, "No-reference blur assessment of digital pictures based on multifeature classifiers," *IEEE Trans. Image Process.*, vol. 20, no. 1, pp. 64–75, Jan. 2011.

[9] M. V. Conde et al., "AIS 2024 challenge on video quality assessment of user-generated content: Methods and results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 5826–5837.

[10] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," 2023, *arXiv:2305.06500*.

[11] D. De, S. Mitra, and R. Soundararajan, "GenzIQA: Generalized image quality assessment using prompt-guided latent diffusion models," 2024, *arXiv:2406.04654*.

[12] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2020.

[13] A. Grattafiori et al., "The llama 3 herd of models," 2024, *arXiv:2407.21783*.

[14] Y. Fang, H. Zhu, Y. Zeng, K. Ma, and Z. Wang, "Perceptual quality assessment of smartphone photography," in *Proc. Int. Conf. Comput. Vis.*, Oct. 2020, pp. 3677–3686.

[15] C. Feichtenhofer, H. Fan, J. Malik, and K. He, "SlowFast networks for video recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6201–6210.

[16] Y. Gao, X. Min, Y. Cao, X. Liu, and G. Zhai, "No-reference image quality assessment: Obtain MOS from image quality score distribution," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1840–1854, Feb. 2025.

[17] D. Ghadiyaram and A. C. Bovik, "Massive online crowdsourced study of subjective and objective picture quality," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 372–387, Jan. 2015.

[18] D. Ghadiyaram, J. Pan, A. C. Bovik, A. K. Moorthy, P. Panda, and K.-C. Yang, "In-capture mobile video distortions: A study of subjective behavior and objective algorithms," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 9, pp. 2061–2077, Sep. 2018.

[19] K. Gu, G. Zhai, W. Lin, and M. Liu, "The analysis of image contrast: From quality assessment to automatic enhancement," *IEEE Trans. Cybern.*, vol. 46, no. 1, pp. 284–297, Jan. 2016.

[20] I. P. Gunawan and M. Ghanbari, "Reduced-reference video quality assessment using discriminative local harmonic strength with motion consideration," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 18, no. 1, pp. 71–83, Jan. 2008.

[21] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[22] V. Hosu et al., "The Konstanz natural video database (KoNViD-1k)," in *Proc. IEEE Int. Conf. Qual. Multimedia Exper.*, 2017, pp. 1–6.

[23] V. Hosu, H. Lin, T. Sziranyi, and D. Saupe, "KonIQ-10k: An ecologically valid database for deep learning of blind image quality assessment," *IEEE Trans. Image Process.*, vol. 29, pp. 4041–4056, 2020.

[24] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, "Perceiver: General perception with iterative attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 4651–4664.

[25] J. Kaplan et al., "Scaling laws for neural language models," 2020, *arXiv:2001.08361*.

[26] J. Korhonen, "Two-level approach for no-reference consumer video quality assessment," *IEEE Trans. Image Process.*, vol. 28, pp. 5923–5938, 2019.

[27] B. Li, W. Zhang, M. Tian, G. Zhai, and X. Wang, "Blindly assess quality of in-the-wild videos via quality-aware pre-training and motion perception," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 5944–5958, Sep. 2022.

[28] D. Li, T. Jiang, and M. Jiang, "Quality assessment of in-the-wild videos," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 2351–2359.

Authorized licensed use limited to: De La Salle University. Downloaded on February 24,2026 at 21:48:46 UTC from IEEE Xplore. Restrictions apply.

[29] F. Li et al., "LLaVA-NeXT-interleave: Tackling multi-image, video, and 3D in large multimodal models," 2024, *arXiv:2407.07895*.
[30] K. C. Li et al., "Videochat: Chat-centric video understanding," 2024, *arXiv:2305.06355*.
[31] Y. Li, C. Wang, and J. Jia, "LLaMA-VID: An image is worth 2 tokens in large language models," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2024, pp. 323–340.
[32] B. Lin et al., "Video-LLaVA: Learning united visual representation by alignment before projection," 2023, *arXiv:2311.10122*.
[33] J. Lin, H. Yin, W. Ping, P. Molchanov, M. Shoeybi, and S. Han, "VILA: On pre-training for visual language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26679–26689.
[34] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26286–26296.
[35] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2023, pp. 34892–34916.
[36] H. Liu et al., "Ada-DQA: Adaptive diverse quality-aware feature acquisition for video quality assessment," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 6695–6704.
[37] R. Liu, C. Li, Y. Ge, T. H. Li, Y. Shan, and G. Li, "BT-adapter: Video conversation is feasible without video instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13658–13667.
[38] W. Liu, Z. Duanmu, and Z. Wang, "End-to-end blind quality assessment of compressed videos using deep neural networks," in *Proc. 26th ACM Int. Conf. Multimedia*, Oct. 2018, pp. 546–554.
[39] Z. Liu et al., "Video Swin transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 3202–3211.
[40] M. Maaz, H. Rasheed, S. Khan, and F. Khan, "Video-ChatGPT: Towards detailed video understanding via large vision and language models," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics Long Papers*, vol. 1, 2024, pp. 12585–12602.
[41] K. Manasa and S. S. Channappayya, "An optical flow-based full reference video quality assessment algorithm," *IEEE Trans. Image Process.*, vol. 25, no. 6, pp. 2480–2492, Jun. 2016.
[42] X. Min, H. Duan, W. Sun, Y. Zhu, and G. Zhai, "Perceptual video quality assessment: A survey," *Sci. China Inf. Sci.*, vol. 67, no. 11, Nov. 2024, Art. no. 211301.
[43] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.
[44] A. Mittal, M. A. Saad, and A. C. Bovik, "A completely blind video integrity Oracle," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 289–300, Jan. 2015.
[45] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'completely blind' image quality analyzer," *IEEE Signal Process. Lett.*, vol. 20, no. 3, pp. 209–212, Apr. 2012.
[46] V. Ordóñez, G. Kulkarni, and T. L. Berg, "Im2Text: Describing images using 1 million captioned photographs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 24, Dec. 2011, pp. 1143–1151.
[47] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 27730–27744.
[48] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
[49] M. A. Saad, A. C. Bovik, and C. Charrier, "Blind image quality assessment: A natural scene statistics approach in the DCT domain," *IEEE Trans. Image Process.*, vol. 21, no. 8, pp. 3339–3352, Aug. 2012.
[50] M. A. Saad, A. C. Bovik, and C. Charrier, "Blind prediction of natural video quality," *IEEE Trans. Image Process.*, vol. 23, pp. 1352–1365, 2014.
[51] C. Schuhmann et al., "LAION-400M: Open dataset of CLIP-filtered 400 million image-text pairs," in *Proc. NeurIPS Workshop Datacentric AI, DCAI*, Dec. 2021, p.5.
[52] X. Shang, J. Liang, G. Wang, H. Zhao, C. Wu, and C. Lin, "Color-sensitivity-based combined PSNR for objective video quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 5, pp. 1239–1250, May 2019.
[53] P. Sharma, N. Ding, S. Goodman, and R. Soricut, "Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics*, 2018, pp. 2556–2565.
[54] Z. Sinno and A. C. Bovik, "Large-scale study of perceptual video quality," *IEEE Trans. Image Process.*, vol. 28, no. 2, pp. 612–627, Feb. 2019.
[55] R. Soundararajan and A. C. Bovik, "Video quality assessment by reduced reference spatio-temporal entropic differencing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 23, no. 4, pp. 684–694, Apr. 2012.
[56] W. Sun, X. Min, W. Lu, and G. Zhai, "A deep learning based no-reference quality assessment model for UGC videos," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 856–865.
[57] W. Sun, T. Wang, X. Min, F. Yi, and G. Zhai, "Deep learning based full-reference and no-reference quality assessment models for compressed UGC videos," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2021, pp. 1–6.
[58] W. Sun, W. Wen, X. Min, L. Lan, G. Zhai, and K. Ma, "Analysis of video quality datasets via design of minimalistic video quality models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 11, pp. 7056–7071, Nov. 2024.
[59] W. Sun et al., "Enhancing blind video quality assessment with rich quality-aware features," 2024, *arXiv:2405.08745*.
[60] J. S. Pérez, E. Meinhardt-Llopis, and G. Facciolo, "Tv-L1 optical flow estimation," *Image Process. Line*, vol. 3, pp. 137–150, Jan. 2013, doi: 10.5201/IPOL.2013.26.
[61] B. Thomee et al., "YFCC100M: The new data in multimedia research," *Commun. ACM*, vol. 59, no. 2, pp. 64–73, Jan. 2016.
[62] Z. Tu, Y. Wang, N. Birkbeck, B. Adsumilli, and A. C. Bovik, "UGC-VQA: Benchmarking blind video quality assessment for user generated content," *IEEE Trans. Image Process.*, vol. 30, pp. 4449–4464, 2021.
[63] Z. Tu, X. Yu, Y. Wang, N. Birkbeck, B. Adsumilli, and A. C. Bovik, "RAPIQUE: Rapid and accurate video quality prediction of user generated content," *IEEE Open J. Signal Process.*, vol. 2, pp. 425–440, 2021.
[64] Y. Wang, S. Inguva, and B. Adsumilli, "YouTube UGC dataset for video compression research," in *Proc. IEEE 21st Int. Workshop Multimedia Signal Process. (MMSP)*, Sep. 2019, pp. 1–5.
[65] Y. Wang et al., "Rich features for perceptual quality assessment of UGC videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13430–13439.
[66] Y. Wang et al., "Self-instruct: Aligning language models with self-generated instructions," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguistics Long Papers*, vol. 1, 2023, pp. 13484–13508.
[67] H. Wu et al., "Fast-VQA: Efficient end-to-end video quality assessment with fragment sampling," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland. Springer, 2022, pp. 538–554.
[68] H. Wu et al., "DisCoVQA: Temporal distortion-content transformers for video quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 9, pp. 4840–4854, Sep. 2023.
[69] H. Wu et al., "Exploring video quality assessment on user generated contents from aesthetic and technical perspectives," in *Proc. Int. Conf. Comput. Vis.*, Oct. 2023, pp. 20144–20154.
[70] H. Wu et al., "Q-bench: A benchmark for general-purpose foundation models on low-level vision," in *Proc. 12th Int. Conf. Learn. Represent.*, Jan. 2023, pp. 10404–10418.
[71] H. Wu et al., "Q-instruct: Improving low-level visual abilities for multi-modality foundation models," 2023, *arXiv:2311.06783*.
[72] H. Wu et al., "Q-align: Teaching LMMs for visual scoring via discrete text-defined levels," in *Proc. 41st Int. Conf. Mach. Learn.*, Jan. 2023, pp. 54015–54029.
[73] F. Xing, M. Li, Y.-G. Wang, G. Zhu, and X. Cao, "CLIPVQA: Video quality assessment via CLIP," 2024, *arXiv:2407.04928*.
[74] Q. Ye et al., "MPLUG-OwI2: Revolutionizing multi-modal large language model with modality collaboration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13040–13051.
[75] F. Yi, M. Chen, W. Sun, X. Min, Y. Tian, and G. Zhai, "Attention based network for no-reference UGC video quality assessment," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 1414–1418.
[76] Z. Ying, M. Mandal, D. Ghadiyaram, and A. Bovik, "Patch-VQ: 'Patching up' the video quality problem," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14014–14024.
[77] Z. Ying, H. Niu, P. Gupta, D. Mahajan, D. Ghadiyaram, and A. Bovik, "From patches to pictures (PaQ-2-PiQ): Mapping the perceptual space of picture quality," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 3575–3585.
[78] X. Yu et al., "Subjective quality assessment of user-generated content gaming videos," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2022, pp. 74–83.
[79] K. Yuan et al., "PTM-VQA: Efficient video quality assessment leveraging diverse PreTrained models from the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2835–2845.

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[80] H. Zhang, X. Li, and L. Bing, "Video-LLaMA: An instruction-tuned audio-visual language model for video understanding," in *Proc. Conf. Empirical Methods Natural Lang. Process., Syst. Demonstrations*, 2023, pp. 543–553.
[81] P. Zhang et al., "InternLM-XComposer: A vision-language large model for advanced text-image comprehension and composition," 2023, *arXiv:2309.15112*.
[82] R. Zhang et al., "LLaMA-adapter: Efficient fine-tuning of language models with zero-init attention," 2023, *arXiv:2303.16199*.
[83] W. Zhang, K. Ma, G. Zhai, and X. Yang, "Uncertainty-aware blind image quality assessment in the laboratory and wild," *IEEE Trans. Image Process.*, vol. 30, pp. 3474–3486, 2021.
[84] Z. Zhang, H. Wu, E. Zhang, G. Zhai, and W. Lin, "A benchmark for multi-modal foundation models on low-level vision: From single images to pairs," 2024, *arXiv:2402.07116*.
[85] Z. Zhang et al., "MD-VQA: Multi-dimensional quality assessment for UGC live videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 1746–1755.
[86] H. Zhu, B. Chen, L. Zhu, and S. Wang, "Learning spatiotemporal interactions for user-generated video quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1031–1042, Mar. 2022.
[87] H. Zhu et al., "2AFC prompting of large multimodal models for image quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12873–12878, Dec. 2024.
[88] K. Zhu, C. Li, V. Asari, and D. Saupe, "No-reference video quality assessment based on artifact measurement and statistical analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 4, pp. 533–546, Apr. 2015.

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